

2) Evaluate $x^2 - 5(x - y)$ for $x = 6$ and $y = 2$

$$\begin{aligned}
 [x]^2 - 5([x] - [y]) &= [6]^2 - 5([6] - [2]) \\
 &= 36 - 5(6 - 2) \\
 &= 36 - 5(4) \\
 &= 36 - 20 \\
 &= \mathbf{16}
 \end{aligned}$$

24) Evaluate $(-3)^3(-2)^2$

$$\begin{aligned}
 (-3)^3(-2)^2 &= (-3)(-3)(-3)(-2)(-2) \\
 &= -27 \cdot 4 \\
 &= \mathbf{-108}
 \end{aligned}$$

30) Evaluate $(2x^3)^{-4}$

$$\begin{aligned}
 (2x^3)^{-4} &= 2^{-4}x^{3 \cdot (-4)} \\
 &= \frac{1}{2^4}x^{-12} \\
 &= \frac{1}{2^4x^{12}} \\
 &= \mathbf{\frac{1}{16x^{12}}}
 \end{aligned}$$

32) Evaluate 3.74×10^4

$$\begin{aligned}
 &= 37.4 \times 10^3 \\
 &= 374 \times 10^2 \\
 &= 3,740 \times 10^1 \\
 &= \mathbf{37,400}
 \end{aligned}$$

32) Simplify $\frac{\sqrt{96x^3}}{\sqrt{2x}}$

$$\begin{aligned}
 &= \sqrt{\frac{96x^3}{2x}} \\
 &= \sqrt{48x^2} \\
 &= x\sqrt{2^4 \cdot 3} \\
 &= 2^2x\sqrt{3} \\
 &= 4x\sqrt{3}
 \end{aligned}$$

90) Factor $x^2 - 11x + 28$

$$\begin{aligned}
 (-7)(-4) &= 28 \text{ and } -7x + -4x = -11x \\
 \text{so } x^2 - 11x + 28 &= (x - 7)(x - 4)
 \end{aligned}$$

98) Factor $x^4 - 16$

$$\begin{aligned}
 \text{The difference of squares: } a^2 - b^2 &= (a + b)(a - b) \\
 x^4 - 16 &= (x^2 - 4)(x^2 + 4) \\
 &= (\mathbf{x - 2})(\mathbf{x + 2})(\mathbf{x^2 + 4})
 \end{aligned}$$

110) Simplify $\frac{x^2+3x-18}{x^2-36}$

$$\begin{aligned}
 &= \frac{x^2+3x-18}{(x-6)(x+6)} \quad \text{Diff. of Squares} \\
 &= \frac{(x+6)(x-3)}{(x-6)(x+6)} \\
 &= \frac{\mathbf{x-3}}{\mathbf{x-6}} \quad \mathbf{x \neq \pm 6}
 \end{aligned}$$

120) Simplify $\frac{3+\frac{12}{x}}{1-\frac{16}{x^2}}$

$$\begin{aligned}
 &= \frac{\frac{3x}{x} + \frac{12}{x}}{\frac{x^2}{x^2} - \frac{16}{x^2}} \\
 &= \frac{\frac{3x+12}{x}}{\frac{x^2-16}{x^2}} \\
 &= \left(\frac{3x+12}{x}\right) \frac{x^2}{x^2-16} \\
 &= \frac{x(3x+12)}{x^2-16} \\
 &= \frac{3x(x+4)}{(x-4)(x+4)} \\
 &= \frac{\mathbf{3x}}{\mathbf{x-4}} \quad \mathbf{x \neq 0, \pm 4}
 \end{aligned}$$

Evaluate

1. $3 + 6(x - 3)^3$ for $x = 4$

2. $5(2x^2 - 6x) - (2 + \frac{8}{x+5})$ for $x = -3$

3. $x^2 - 5x$ for $x = a + 1$

4. $\frac{2x+y}{xy-2x}$ for $x = -2$ and $y = 4$

Simplify each expression

5. $5(3x + 4) - 4$

6. $6 - 5[8 - (2y - 4)]$

7. $(2x^2y^3 - xy + y^2) - (-4x^2y^3 - 5xy - y^2)$

8. $\frac{x}{x+3} + \frac{5}{x-3}$